

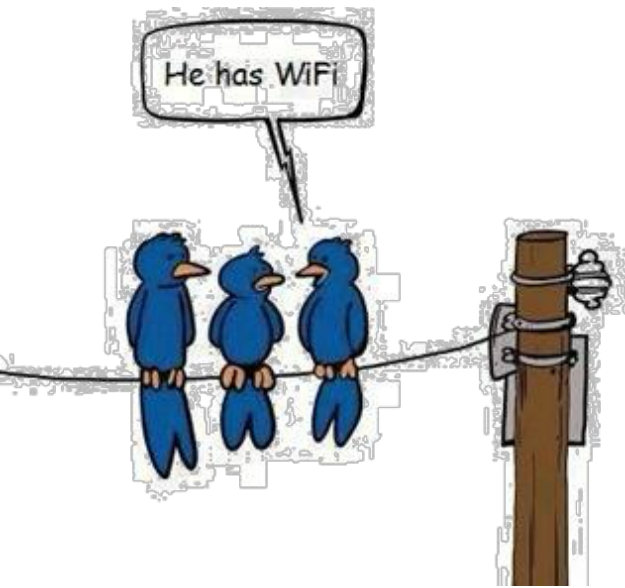
ECSE 4560/ 6560

Modern Communication Systems

Lecture 18: MIMO Part3: STBC

Diversity

Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **STBC Diversity**
- Projects
 - Group formation [MCS Fall25 project.xlsx](#)
 - Project mid-term report due **Friday 11/7**
 - Final Project Presentation and Demos: Dec 9 and Dec 12
 - Project Report and peer evaluation: Finals exam week
- HW/Exams Dates
 - HW5 (MIMO Theory) due **Mon 11/10**. HW6 (MIMO code) due 11/14
 - Exam 1: Friday Oct 24 Graded (Submit regrading request by Tue 11/4)
 - Exam 2: **Friday Nov 21**

Extra Reading on Piazza

- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - https://piazza.com/class_profile/get_resource/mevvk9kirhg2fu/mgb9esiush64ae
 - Parts of Chapter 1,2,3,4,7
 - **MIMO Chapter 6**

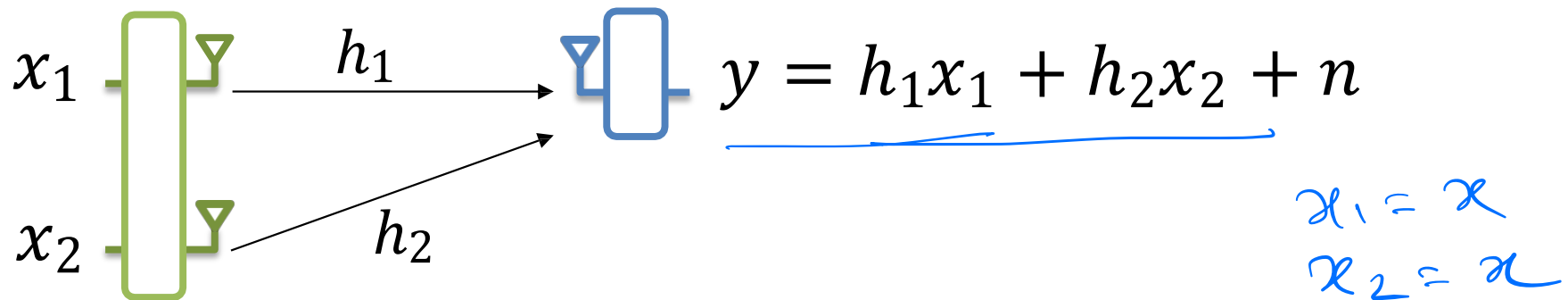
Today's Class

Diversity

- MIMO Gains
 - Receiver Diversity
 - Transmitter Diversity
 - Space-Time codes (Alamouti Codes)
- MIMO Multiplexing Gains
 - Zero-forcing
 - SVD
 - MU MIMO

STBC
for Diversity
without
channel
feedback!

Transmitter Diversity (single carrier or single subcarrier)



What should we transmit on each antenna?

Option 1: transmit the same thing x on both antennas

$$y = (h_1 + h_2)x + n$$

$$h_1 \sim \mathcal{CN}(0, \sigma_h^2)$$

$$h_2 \sim \mathcal{CN}(0, \sigma_h^2)$$

Channels can sum up destructively! $h_1 + h_2 \approx 0$

Total transmit power = $2P \rightarrow$ Doubled TX power

$$h_2 = -h_1$$

$$\Rightarrow (h_1 + -h_1) = 0$$

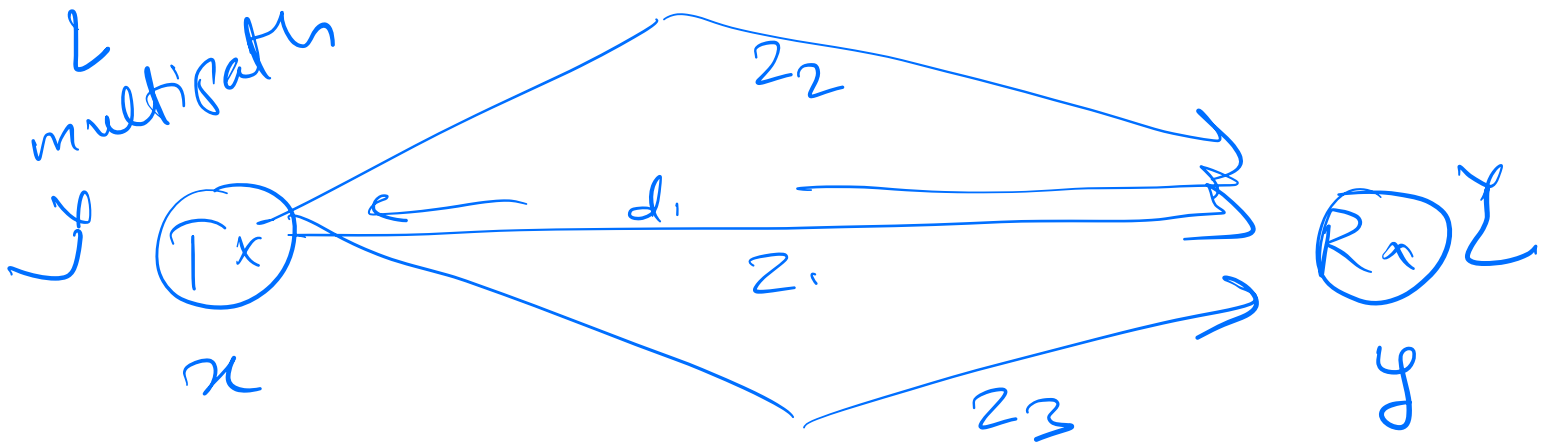
$\rightarrow$ Why not use 1 TX with $2P$

$$h_2 = h_1 \Rightarrow h_1 + h_2 = 2h_1$$

Channel model

$$z_i = \frac{d_i}{c}$$

L multipaths



$$y = \sum_i a_i x(t - z_i) e^{-j2\pi f z_i}$$

convolution

$$y = x(t) \otimes h(t)$$

$$h(t) = \sum_i a_i \delta(t - z_i) e^{-j2\pi f z_i}$$

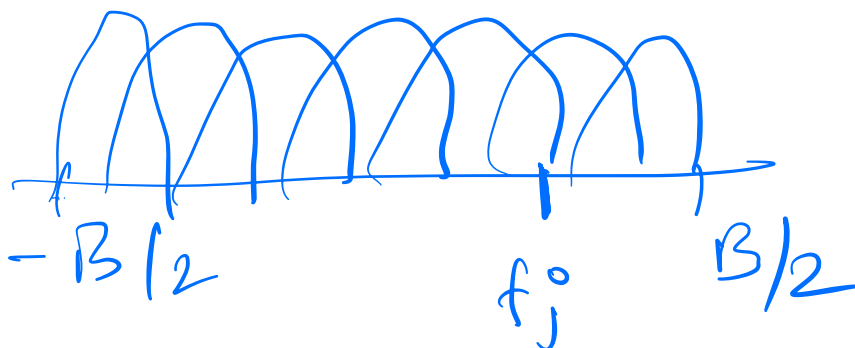
$$e^{-j2\pi f \frac{d_i}{c}}$$

$$f = c/\lambda$$

$$d_i = z_i c$$

$$H(f) = \sum_i a_i e^{-j2\pi f \frac{d_i}{c}} e^{j2\pi f z_i}$$

$$-\frac{B}{2} \text{ to } \frac{B}{2}$$



$$H(f)$$

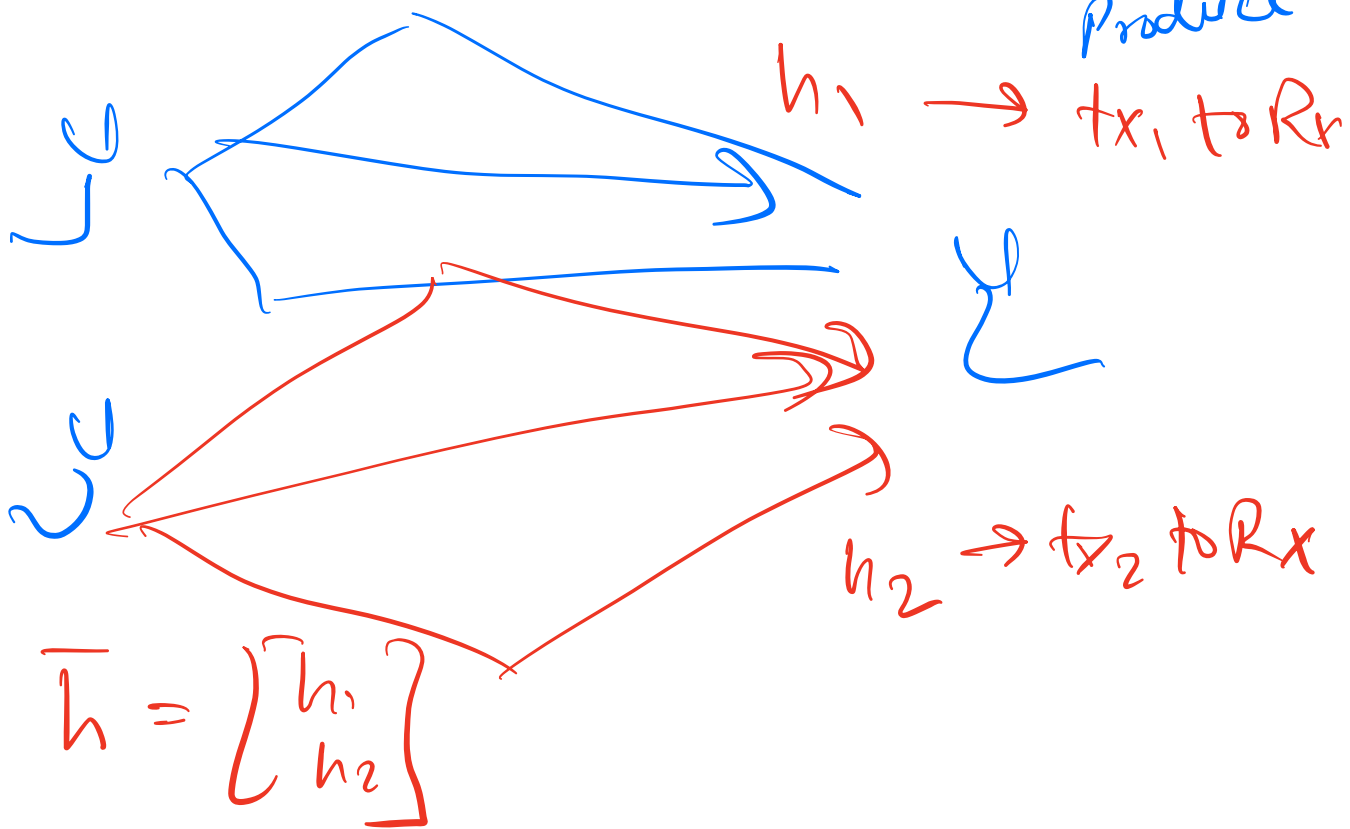
OFDM

transmit
 $x(f_i)$
 subcarrier's
 index

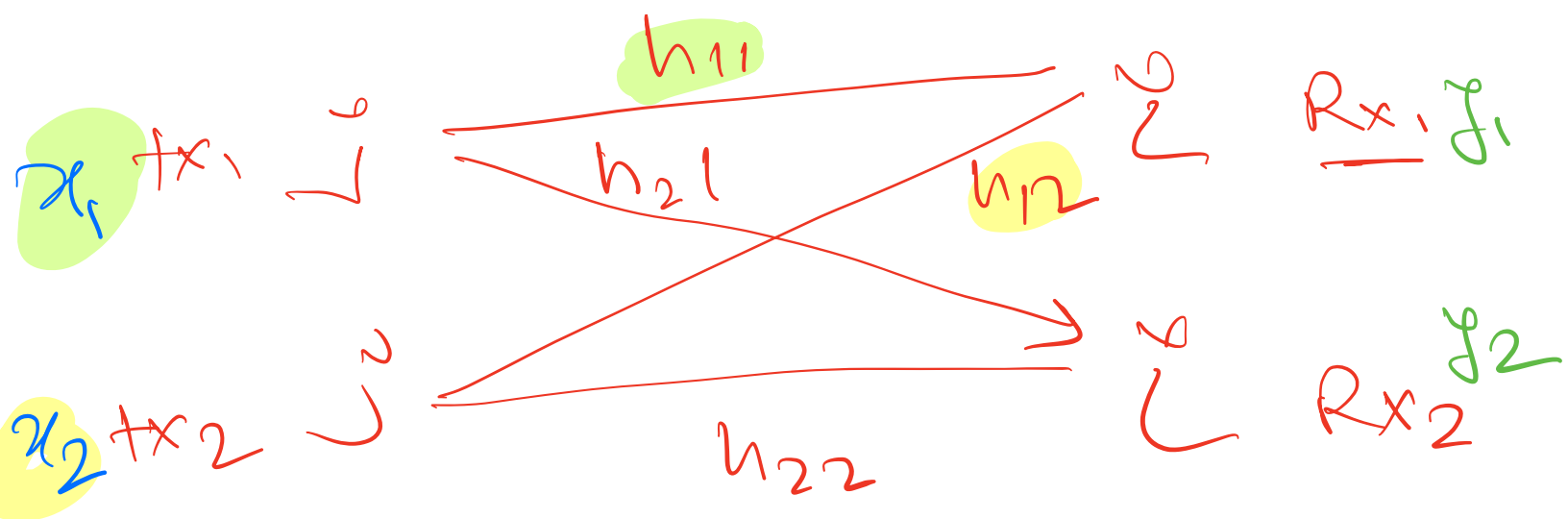
$$\longrightarrow H(f_i) \longrightarrow Y(f_i)$$

$$Y(f_i) = H(f_i) \times x(f_i)$$

product



for each subcarrier 'i'

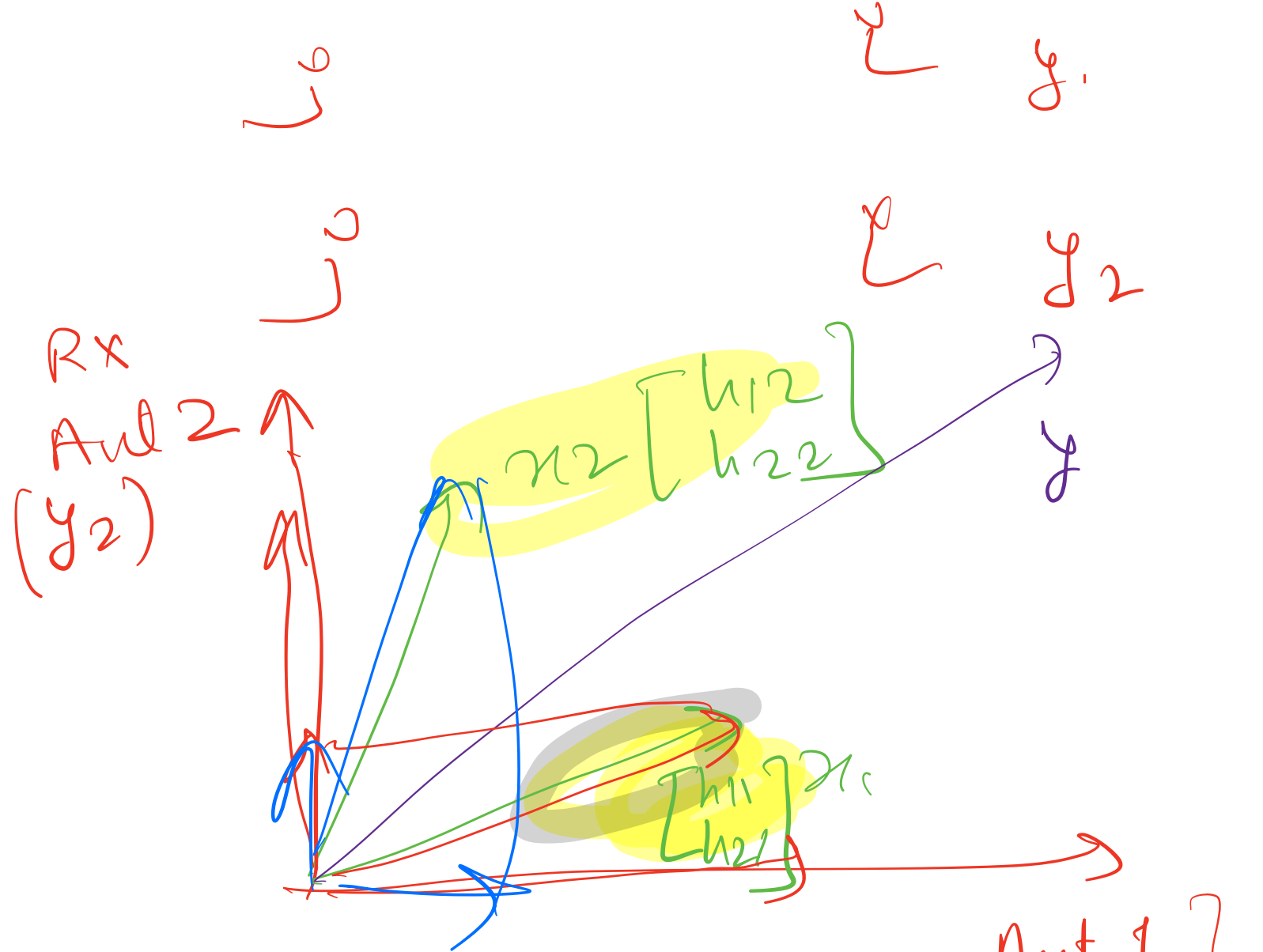


$$\vec{y} = H \vec{x} + \text{noise}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$y_1 = h_{11} x_1 + h_{12} x_2$$

$$y_2 = h_{21} x_1 + h_{22} x_2$$



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} \\ h_{21} \end{bmatrix} x_1$$

$$+ \begin{bmatrix} h_{12} \\ h_{22} \end{bmatrix} x_2$$

Assume everything is real

$$\vec{y} = H \vec{x}$$

$$\vec{x} = H^{-1} \vec{y}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = H^{-1} H \vec{x}$$

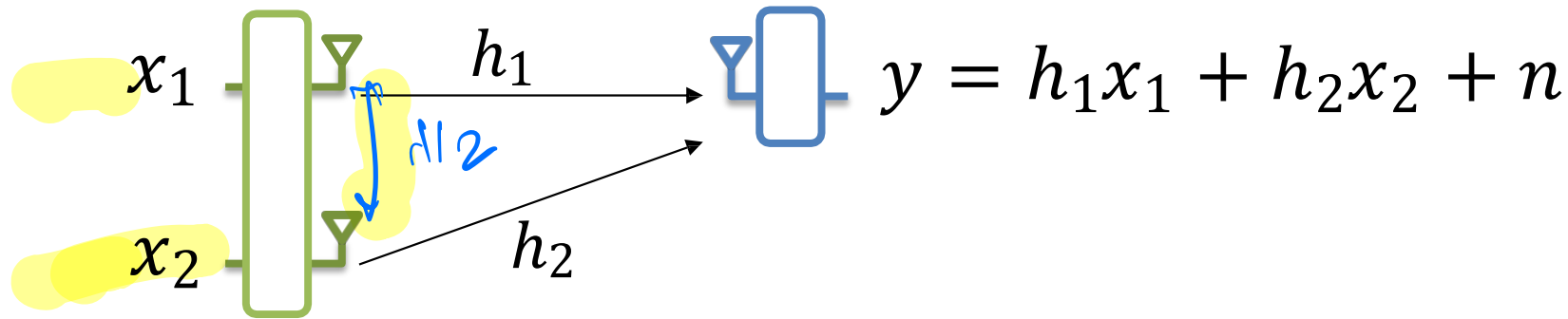
$$= \vec{x}$$

if H is invertible

$$\det(H) \neq 0$$

SVD of $H \Rightarrow$ non zero eigen values

Transmitter Diversity

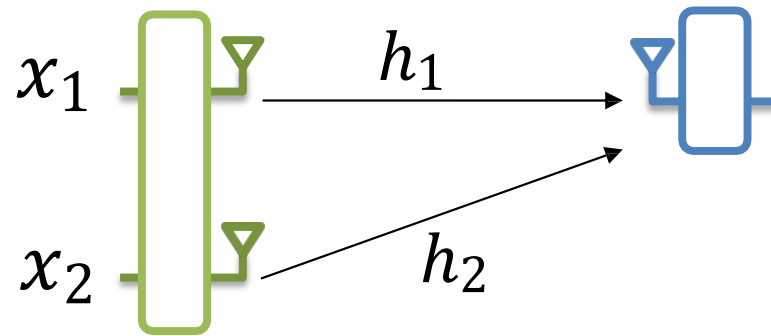


What should we transmit on each antenna?

Option 1: transmit the same thing x on both antennas

- Must ensure signals sum up constructively.
- Must ensure total TX power = $E[|x_1|^2] + E[|x_2|^2]$
 $= E[|x|^2] = P$

Transmitter Diversity



$$y = h_1x_1 + h_2x_2 + n$$

Tx should know h_1 & h_2
(1) Rx can feedback h_i
(2) Tx can estimate from uplink slot
(only works in TDD)

What should we transmit on each antenna?

Option 2: Maximum Ratio Combining (MRC)

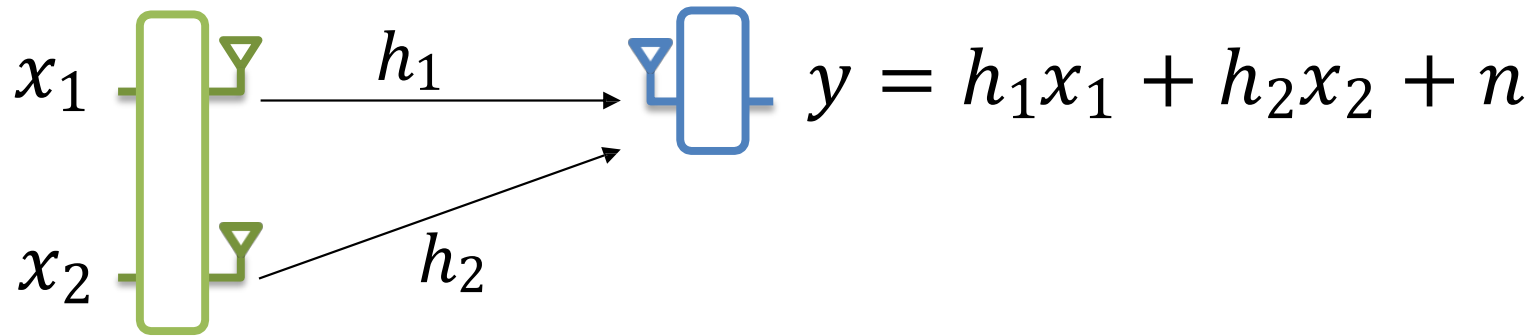
$$\left. \begin{array}{l} x_1 = \alpha_1 x \\ x_2 = \alpha_2 x \end{array} \right\} y = (\alpha_1 h_1 + \alpha_2 h_2)x + n$$

$$\text{Set: } \alpha_1 = h_1^*, \alpha_2 = h_2^* \rightarrow y = (h_1^* h_1 + h_2^* h_2)x + n$$

Power is not normalized

$$= (|h_1|^2 + |h_2|^2)x + n$$

Transmitter Diversity



What should we transmit on each antenna?

Option 2: Maximum Ratio Combining (MRC)

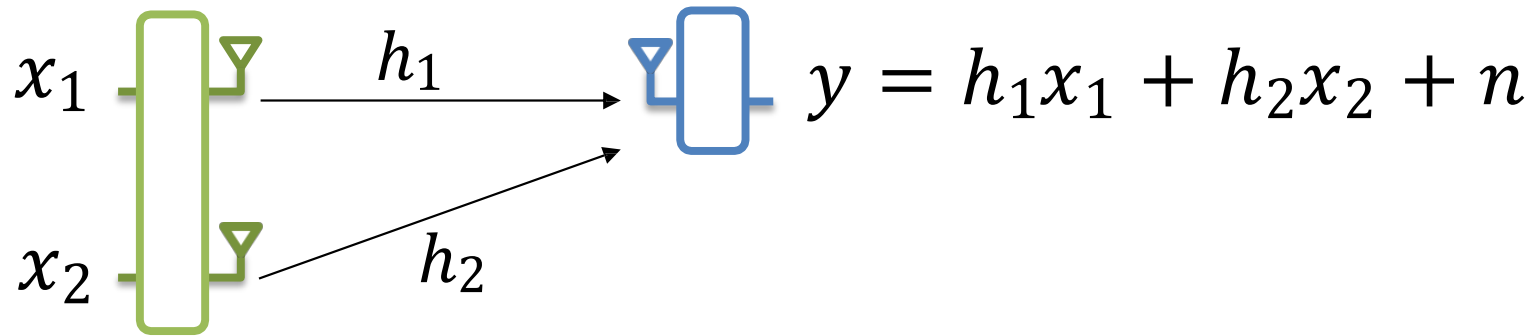
$$x_1 = \frac{h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}} x \quad \leftarrow \text{Power normalized}$$

$$x_2 = \frac{h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}} x$$

Total TX Power = $\mathbb{E}[|x_1|^2] + \mathbb{E}[|x_2|^2]$

$$= \frac{|h_1|^2}{|h_1|^2 + |h_2|^2} P + \frac{|h_2|^2}{|h_1|^2 + |h_2|^2} P = P$$

Transmitter Diversity



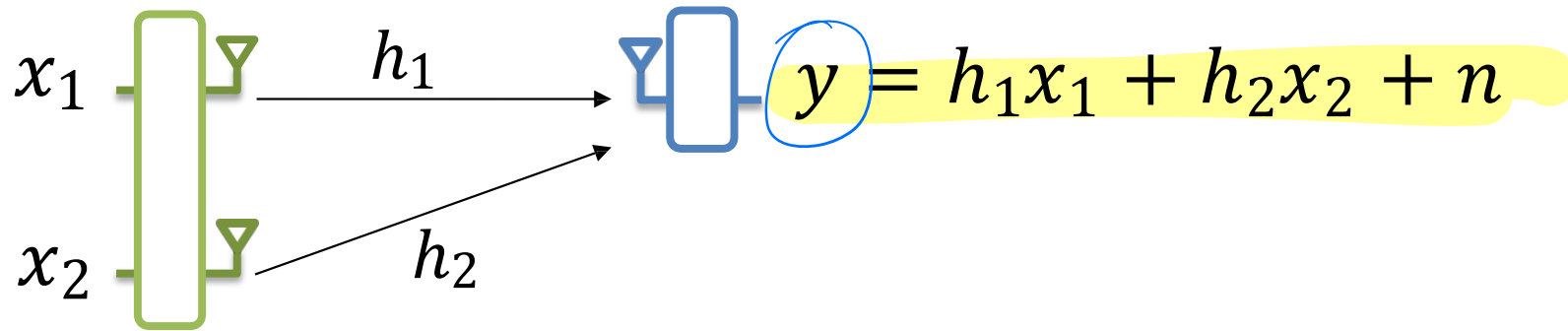
What should we transmit on each antenna?

Option 2: Maximum Ratio Combining (MRC)

$$x_1 = \frac{h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}} x \quad x_2 = \frac{h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}} x$$

$$y = \frac{h_1 h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}} x + \frac{h_2 h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}} x + n$$

Transmitter Diversity



What should we transmit on each antenna?

Option 2: Maximum Ratio Combining (MRC)

$$x_1 = \frac{h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}} x$$

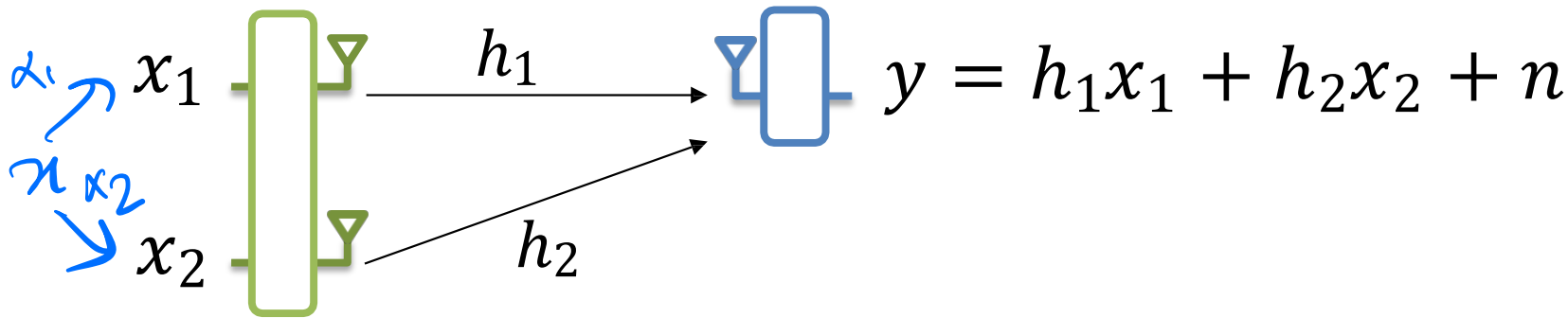
$$x_2 = \frac{h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}} x$$

$$y = \frac{|h_1|^2 + |h_2|^2}{\sqrt{|h_1|^2 + |h_2|^2}} x + n = \left(\sqrt{|h_1|^2 + |h_2|^2} \right) x + n$$

Handwritten note: $\hat{x} = y / \sqrt{|h_1|^2 + |h_2|^2}$

Transmitter Diversity

$$x_1 = a_1 x$$
$$x_2 = a_2 x$$



What should we transmit on each antenna?

Option 2: Maximum Ratio Combining (MRC)

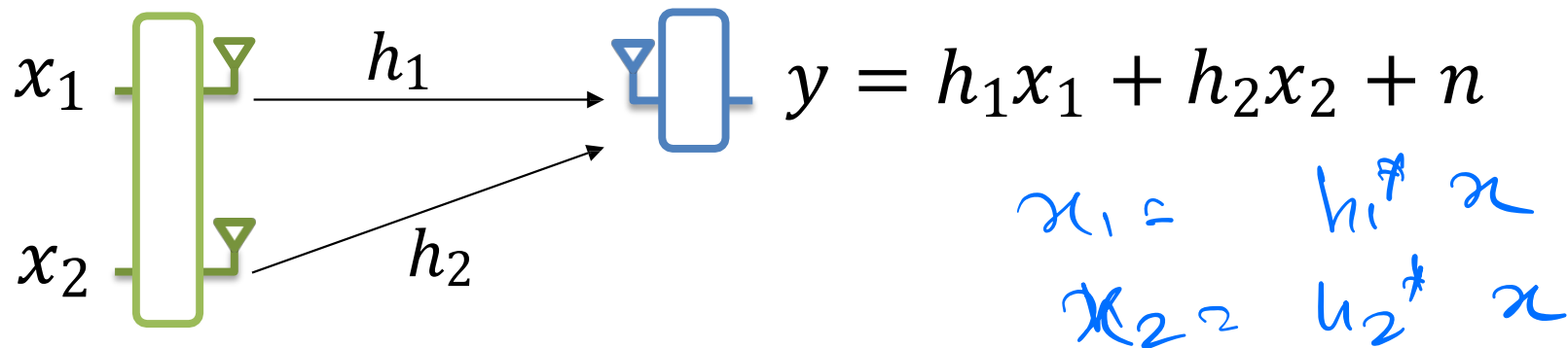
$$y = \left(\sqrt{|h_1|^2 + |h_2|^2} \right) x + n$$

$$P = \mathbb{E} |x|^2$$
$$n \sim \mathcal{CN}(0, \sigma^2)$$

$$SNR = \left(|h_1|^2 + |h_2|^2 \right) \frac{P}{\sigma^2} \rightarrow \text{Similar SNR Gain to RX Diversity}$$

Caveat: MRC at TX Requires Feedback from the Receiver!

Transmitter Diversity – HW5



Consider narrowband flat fading with channel vector

$$\mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} \in \mathbb{C}^2, \quad n \sim \mathcal{CN}(0, N_0), \quad \mathbb{E}[|x|^2] = P.$$

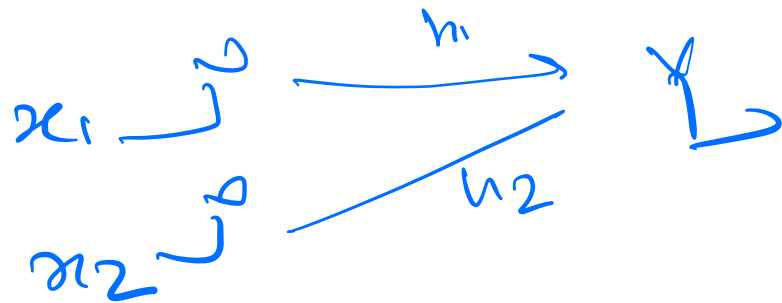
Unless stated otherwise, total TX power per channel use is P (shared across the two antennas).

(a) **MRC beamforming with CSIT (derivation).** With perfect Channel state information at the transmitter (CSIT), the transmitter applies a unit-norm precoder $\mathbf{w} \in \mathbb{C}^2$ and sends $\mathbf{x} = \mathbf{w}x$.

(i) Under $\|\mathbf{w}\|_2 = 1$, derive $\mathbf{w}$ that maximizes instantaneous receive SNR.

(ii) Show that the resulting SNR equals $\gamma_{\text{BF}} = \frac{\|\mathbf{h}\|_2^2 P}{N_0} = \frac{(|h_1|^2 + |h_2|^2)P}{N_0}$.

Transmitter Diversity – HW5



$$y = h_1 x_1 + h_2 x_2 + n$$

$$= \begin{bmatrix} h_1 & h_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + n$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \bar{w} x$$

$$= \bar{h}^T \bar{w} x + n$$

$$SNR = \frac{E | \bar{h}^T \bar{w} x |^2}{E | n |^2} = | \bar{h}^T \bar{w} |^2 \frac{P}{N_0}$$

$$\max | \bar{h}^T \bar{w} |^2 P / N_0$$

$$\text{s.t. } \| w \| = 1$$

Cauchy-Schwarz inequality

Transmitter Diversity – HW5

$$\bar{\omega} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}$$

C.S. inequality

$$|\bar{h}^T \bar{\omega}|^2 \leq \|\bar{h}\|^2 \|\bar{\omega}\|^2$$

$h_1 \alpha_1 + h_2 \alpha_2$
with equality
iff $\bar{\omega} = \bar{h}^*$

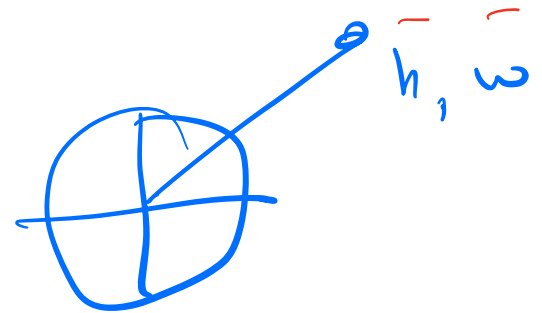
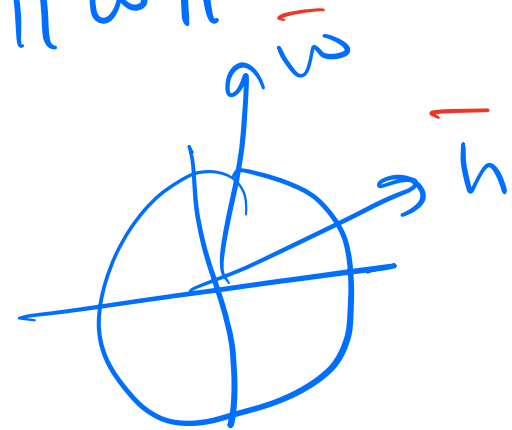
$$\alpha_1 = h_1^*$$

$$\alpha_2 = h_2^*$$

$\|\bar{\omega}\| = 1 \Rightarrow$ normalize

$$\alpha_1 = \frac{h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}}$$

$$\alpha_2 = \frac{h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}}$$



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 - Zero-forcing
 - SVD
 - MU MIMO

N Tx antennas
 $\Rightarrow$ SNR gain N times

selection diversity

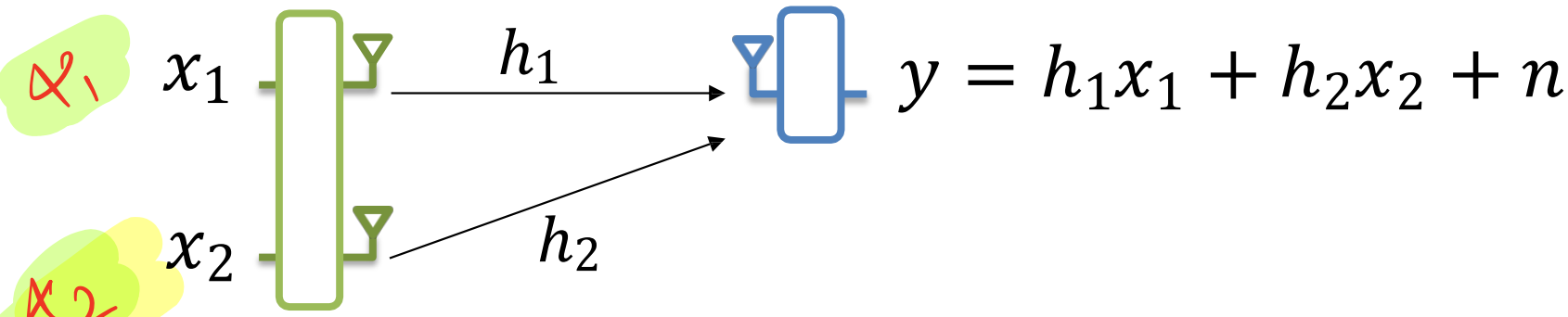
$\Rightarrow$ choose either
 Tx_1 or Tx_2
depending on
which channel
is good

$$SNR \propto \max(|h_1|^2, |h_2|^2) \frac{P}{\sigma^2}$$

STBC

Transmitter Diversity

we have seen this

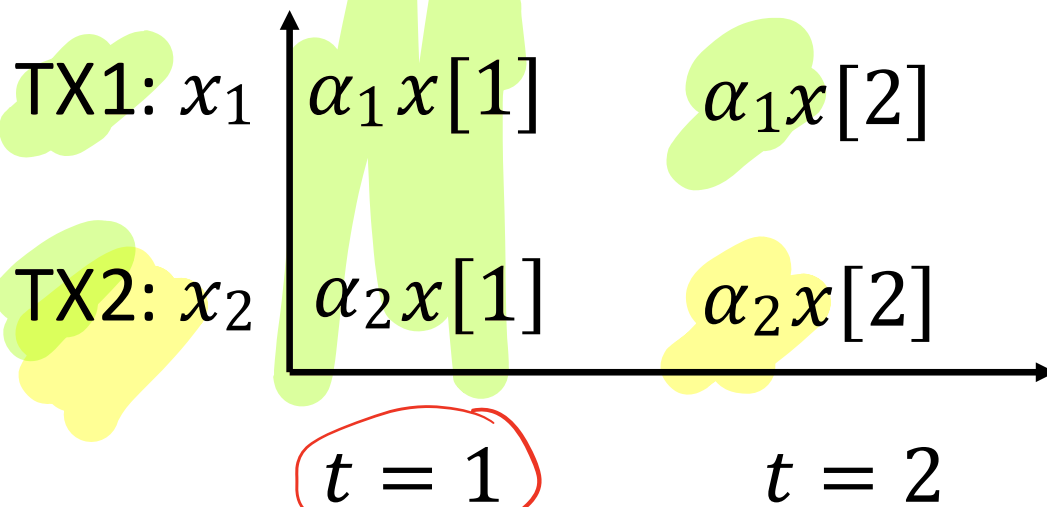


What should we transmit on each antenna?

Solution: Use Space-Time Codes

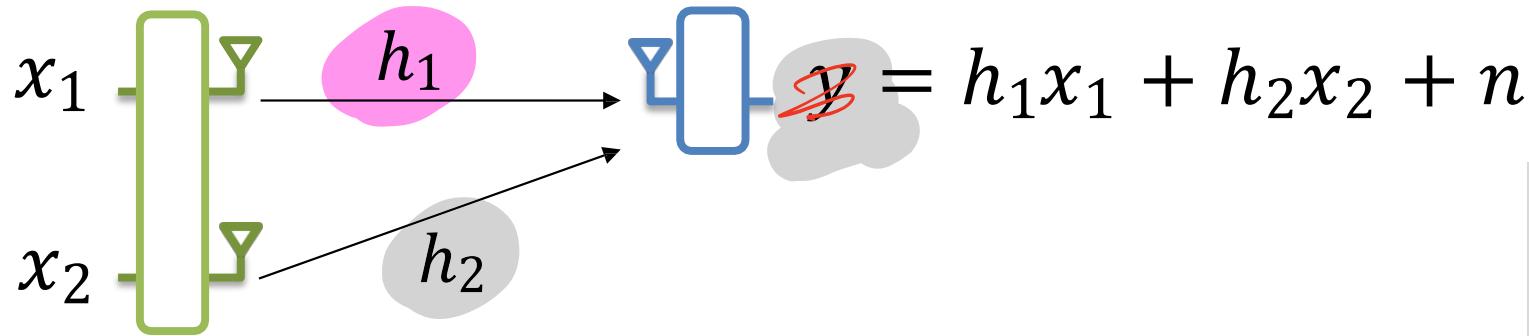
issue

MRC codes across space only (Requires Channel Feedback):



$$\alpha_1 = \frac{h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}}$$
$$\alpha_2 = \frac{h_2^*}{\sqrt{|h_1|^2 + |h_2|^2}}$$

Transmitter Diversity



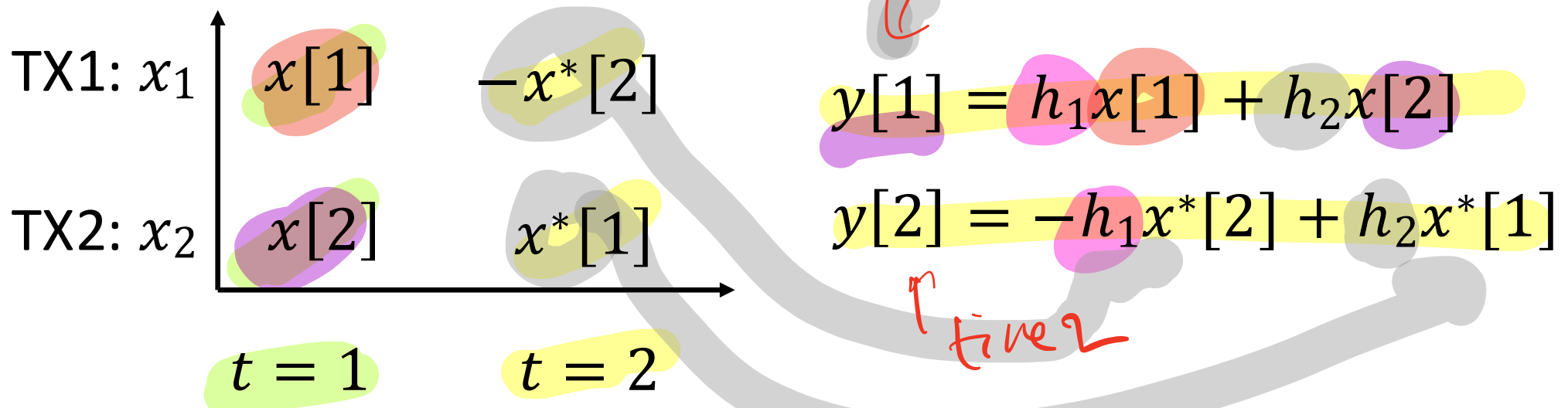
What should we transmit on each antenna?

Solution: Use Space-Time Codes



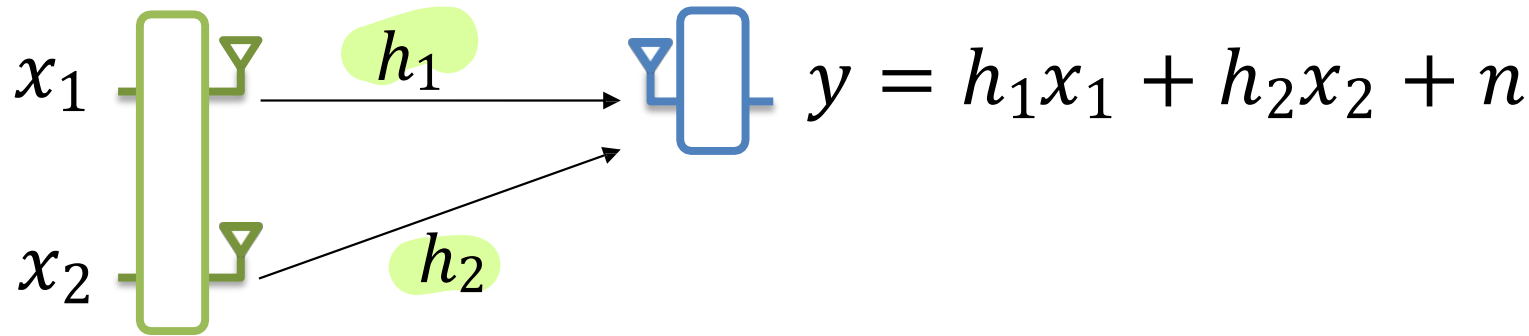
Siavash Alamouti
1996 AT&T Labs

Alamouti Codes:



Transmitter Diversity

Channels don't change in two time slots



What should we transmit on each antenna?

Solution: Use Space-Time Codes

Alamouti Codes:

$$y[1] = h_1x[1] + h_2x[2] \quad y[2] = -h_1x^*[2] + h_2x^*[1]$$

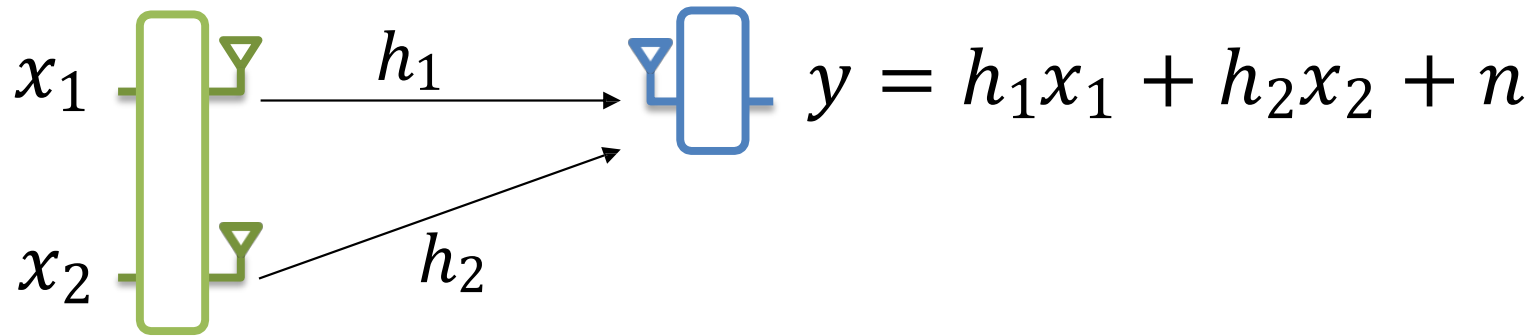
$$h_1^*y[1] + h_2y^*[2] = \underbrace{h_1^*h_1x[1]} + \cancel{h_1^*h_2x[2]} - \cancel{h_2h_1^*x[2]} + h_2h_2^*x[1]$$

$$= (|h_1|^2 + |h_2|^2) \overline{x[1]}$$

magic ↓

removed interf. from $x[2]$

Transmitter Diversity



What should we transmit on each antenna?

Solution: Use Space-Time Codes

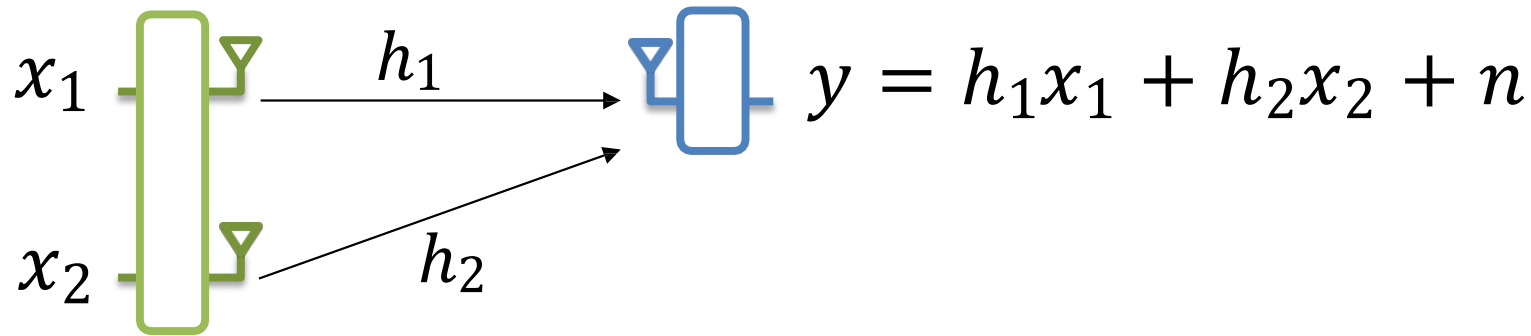
Alamouti Codes:

$$y[1] = h_1x[1] + h_2x[2] \quad y[2] = -h_1x^*[2] + h_2x^*[1]$$

$$\begin{aligned} h_2^*y[1] - h_1y^*[2] &= \cancel{h_2^*h_1x[1]} + h_2^*h_2x[2] \\ &\quad + h_1h_1^*x[2] - \cancel{h_1h_2^*x[1]} \\ &= (|h_1|^2 + |h_2|^2)x[2] \end{aligned}$$

recover $x[2]$
no interf.
from $x[1]$

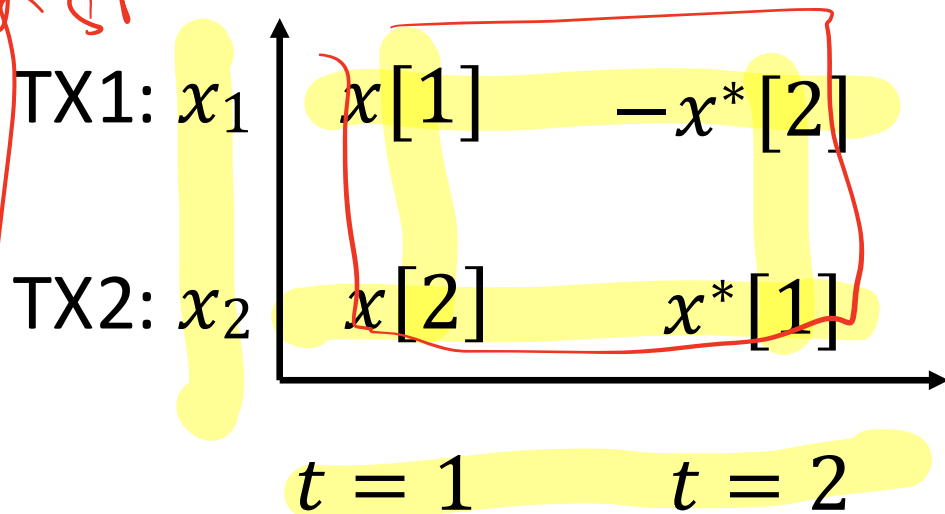
Transmitter Diversity



What should we transmit on each antenna?

Solution: Use Space-Time Codes

Alamouti Codes:



$$h_1^* y[1] + h_2 y^*[2]$$

$$= (|h_1|^2 + |h_2|^2) x[1]$$

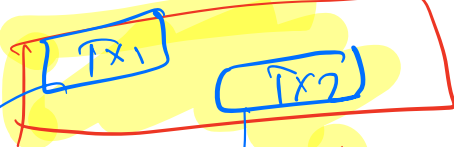
$$h_2^* y[1] - h_1 y^*[2]$$

$$= (|h_1|^2 + |h_2|^2) x[2]$$

not normalized

Preamble

Data



DD DD

mm
↑
channel

ofdm packet

h_1

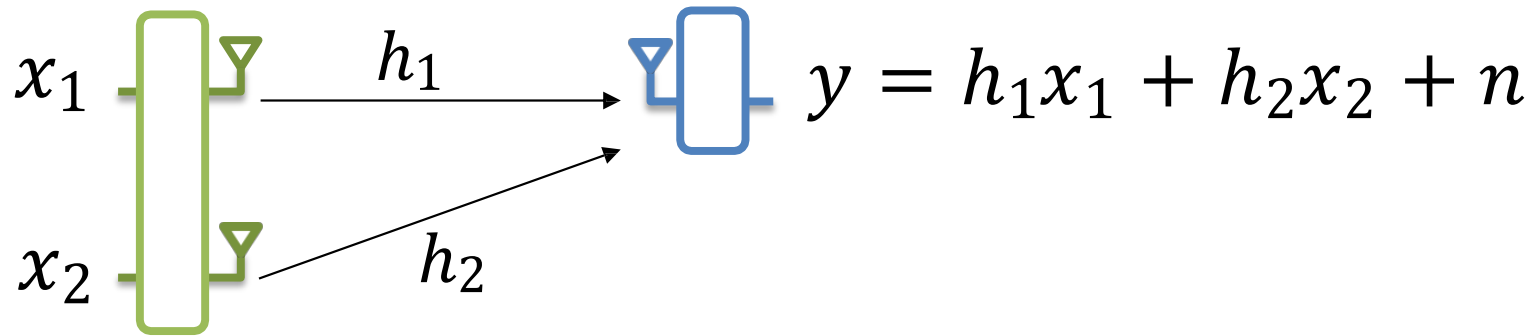
h_2

channel
doesn't
change

squelize
data

single preamble
to estimate
channel

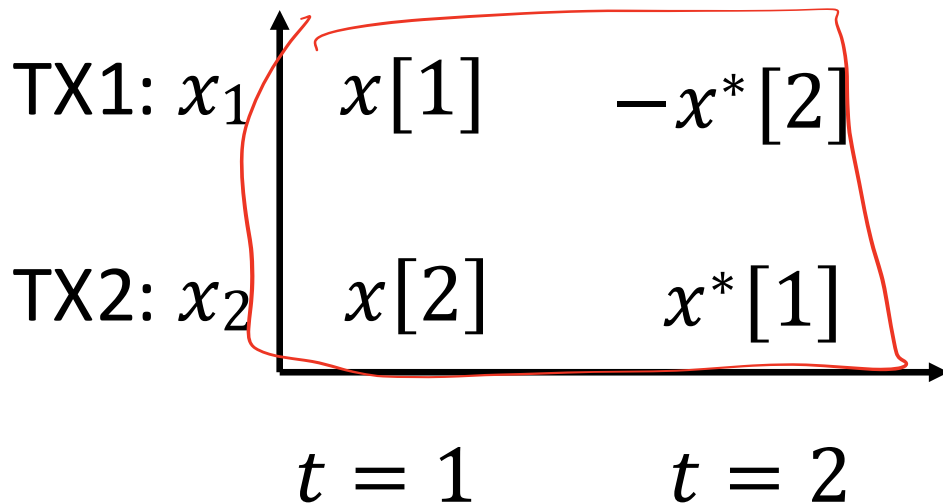
Transmitter Diversity - Normalize



What should we transmit on each antenna?

Solution: Use Space-Time Codes

Alamouti Codes:



$$h_1^* y[1] + h_2 y^*[2]$$

$$= \sqrt{(|h_1|^2 + |h_2|^2)} x[1]$$

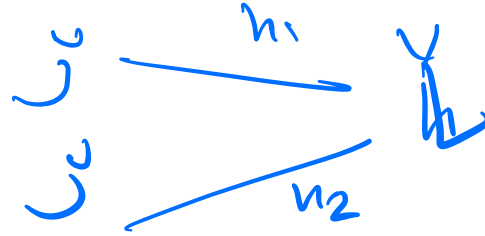
$$h_2^* y[1] - h_1 y^*[2]$$

$$= \sqrt{(|h_1|^2 + |h_2|^2)} x[2]$$

$$\frac{h_2}{\sqrt{|h_1|^2 + |h_2|^2}}$$

after normalization

Alamouti SNR



(c) **Alamouti STBC (no CSIT)**. Using the 2×1 Alamouti code over two time slots:

$$\text{slot } t : [x_1 \ x_2], \quad \text{slot } t + 1 : [-x_2^* \ x_1^*].$$

- (i) Write the two received samples y_t and y_{t+1} in terms of $h_1, h_2, x_1, x_2, n_t, n_{t+1}$.
- (ii) Form the standard two-equation linear combiner that decouples (x_1, x_2) and show that each symbol experiences an effective gain $|h_1|^2 + |h_2|^2$ (assume total TX power per *time slot* equals P and the usual Alamouti normalization).
- (iii) State the resulting per-symbol SNR γ_{Alam} in terms of $P, N_0, |h_1|^2, |h_2|^2$ and the code rate.

Alamouti SNR

$$(|h_1|^2 + |h_2|^2)$$

$$\frac{h_1^*}{\sqrt{|h_1|^2 + |h_2|^2}} y_1 + \frac{h_2}{\sqrt{|h_1|^2 + |h_2|^2}} y_2$$

$$= (|h_1|^2 + |h_2|^2) x_1$$

$$\text{SNR} = \frac{\|\mathbf{h}\|^2}{\sigma_n^2} E\{|x_1|^2\}$$

$$\frac{\|\mathbf{h}^2\|}{\sigma_n^2} P_1, \frac{\|\mathbf{h}^2\|}{\sigma_n^2} P_2, \sqrt{|h_1|^2 + |h_2|^2}$$

$$P_1 = P_2 = \frac{P}{2}$$

$$= \frac{\|\mathbf{h}^2\|}{\sigma_n^2} P_1$$

$$= \sqrt{|h_1|^2 + |h_2|^2} x_1$$

$$\text{SNR} = \frac{(\sqrt{|h_1|^2 + |h_2|^2})^2 E(x_1)^2}{E\{n\}^2}$$

$$\text{SNR} = \frac{P}{2} \frac{\|\mathbf{h}^2\|}{\sigma_n^2} = \frac{1}{2} \frac{\|\mathbf{h}^2\|}{\sigma_n^2} P$$

$$= \frac{(|h_1|^2 + |h_2|^2) P_1}{\sigma_n^2}$$

$$P_1 = \frac{P}{2}$$

SNR for x_1 for time 1

Alamouti SNR across two timeslots

4 terms

SNR for x_1 in time 1

x_2 — 1

x_1 2

x_2 2

same

$$\frac{1}{2} \|h\|^2 \frac{P}{\sigma_n^2}$$

Total SNR / time slot $\Rightarrow \|h\|^2 \frac{P}{\sigma_n^2}$

$\Rightarrow$ same as Tx diversity with CSIT

Code Rate

If an STBC transmits k independent information symbols over T time slots, the **code rate** R is defined as:

$$R = \frac{k}{T} \text{ (symbols per channel use)}$$

2 symbol

2 timeslots

Only the Alamouti code (for 2 transmit antennas) achieves a full code rate of 1 while maintaining full orthogonality and full diversity in complex constellations.

any other STBC code rate < 1

MIMO Gains

very imp!

Multiplexing Gain:

- Send multiple packets at the same time
- $N \times N$ MIMO $\rightarrow \times N$ more packets
- Data Rate: $\propto N \log(SNR/N)$

Diversity Gain:

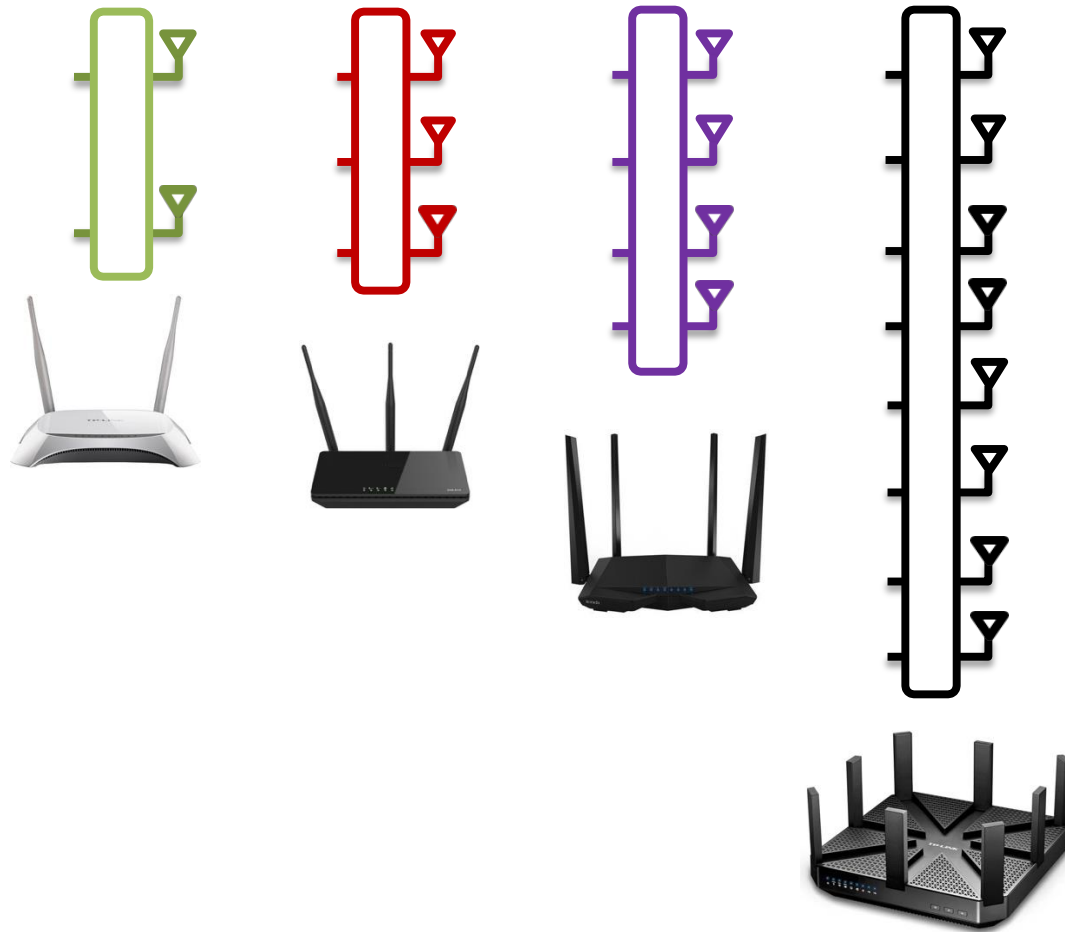
- Increase SNR of the received packets
- $N \times N$ MIMO $\rightarrow \times \log N^2$ data rate
- Data Rate $\propto \log(SNR \times N^2)$

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- **MIMO Multiplexing Gains**
 - **Zero-forcing**
 - **SVD**
 - **MU MIMO**

MIMO: Multiple TX-RX streams

WiFi Access Points 2 to 8 MIMO

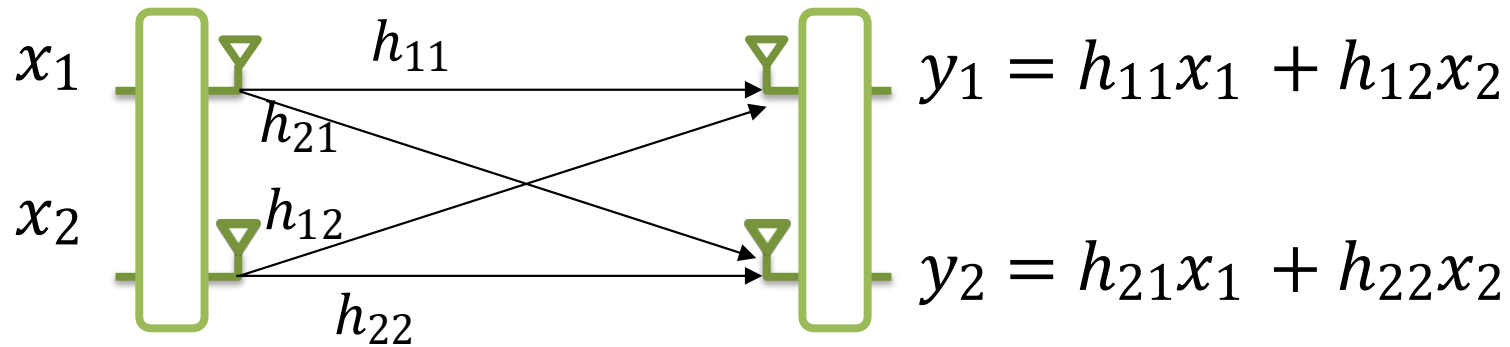


WiFi Devices 1 to 2 MIMO



- Power
- Form Factor & Antenna Separation
- Most mobile phones support 2 MIMO

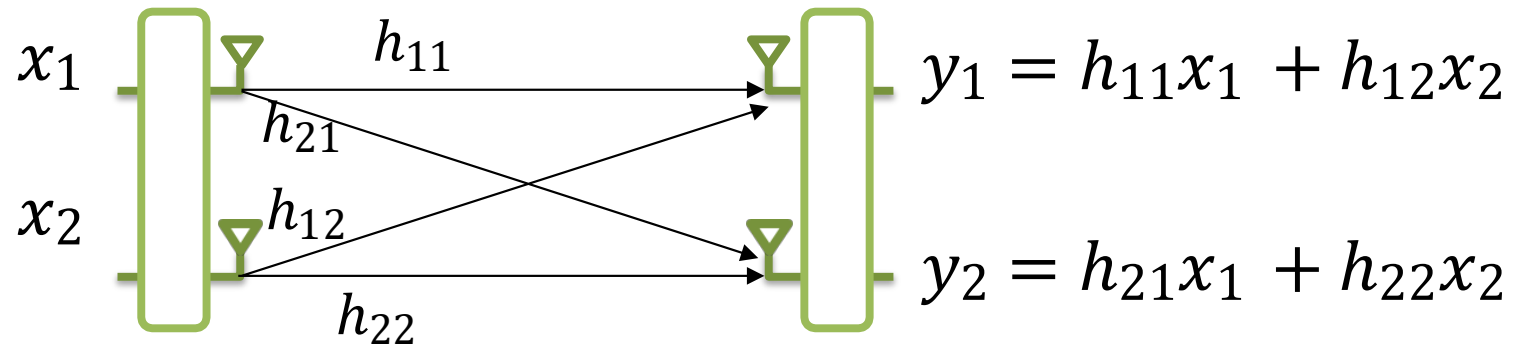
MIMO



$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

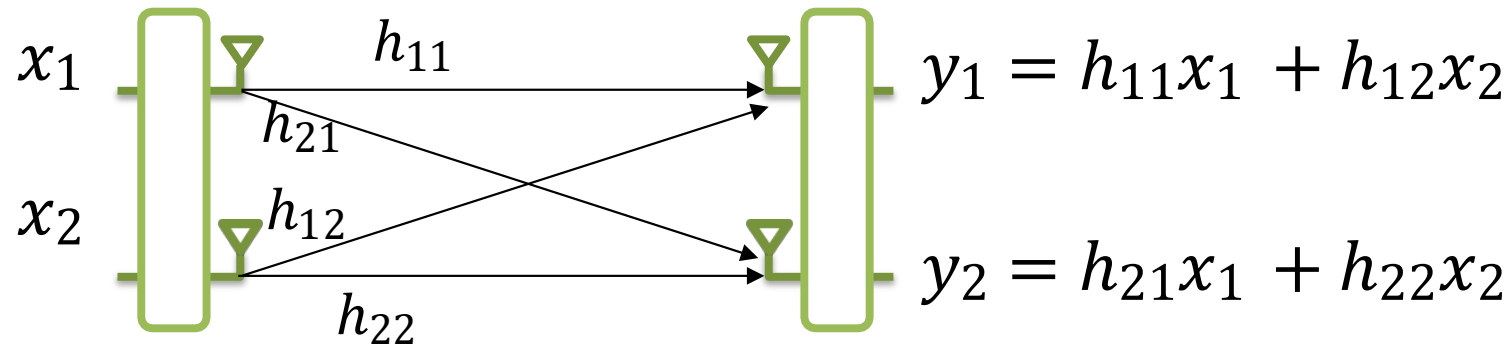
$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

Zero-forcing based MIMO



$$\hat{\mathbf{x}} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{y}$$

SVD-based MIMO



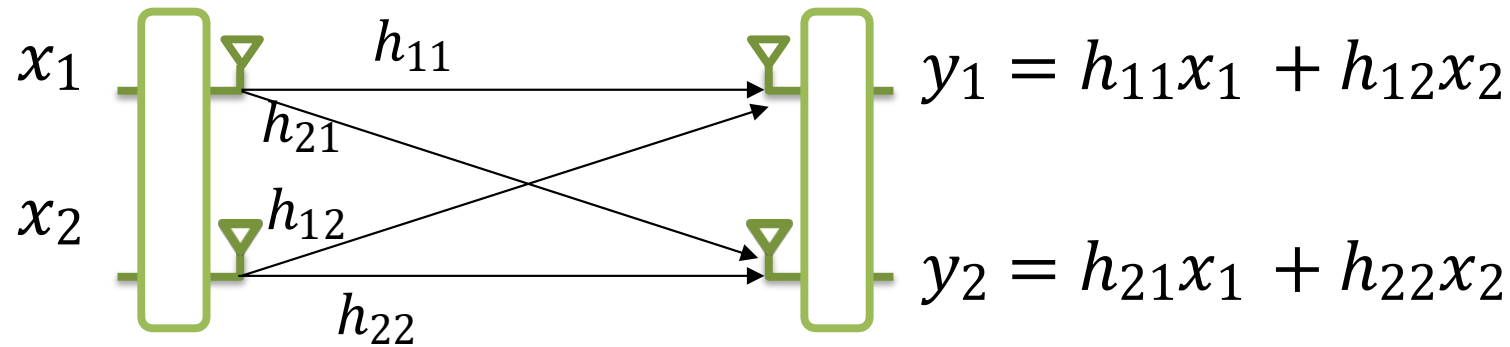
$$\mathbf{H} = \underbrace{\begin{bmatrix} u_{11} & u_{12} & \dots & u_{1t} \\ u_{21} & u_{22} & \dots & u_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ \underbrace{u_{r1}}_{\mathbf{u}_1} & \underbrace{u_{r2}}_{\mathbf{u}_2} & \dots & \underbrace{u_{rt}}_{\mathbf{u}_t} \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix}}_{\mathbf{\Sigma}}$$

$$\underbrace{\begin{bmatrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \\ v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{bmatrix}}_{\mathbf{V}^H},$$

$\left. \begin{matrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \end{matrix} \right\} \mathbf{v}_1^H$
 $\left. \begin{matrix} v_{12}^* & v_{22}^* & \dots & v_{t2}^* \end{matrix} \right\} \mathbf{v}_2^H$
 $\left. \begin{matrix} v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{matrix} \right\} \mathbf{v}_t^H$

$$= \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$$

SVD-based MIMO



$$\mathbf{H} = \underbrace{\begin{bmatrix} u_{11} & u_{12} & \dots & u_{1t} \\ u_{21} & u_{22} & \dots & u_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ \underbrace{u_{r1}}_{\mathbf{u}_1} & \underbrace{u_{r2}}_{\mathbf{u}_2} & \dots & \underbrace{u_{rt}}_{\mathbf{u}_t} \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix}}_{\mathbf{\Sigma}}$$

$$\|\mathbf{u}_i\|^2 = \|\mathbf{v}_i\|^2 = 1, 1 \leq i \leq t$$

$$\mathbf{u}_i^H \mathbf{u}_j = \mathbf{v}_i^H \mathbf{v}_j = 0, i \neq j, 1 \leq i, j \leq t$$

$$\mathbf{V}^H \mathbf{V} = \mathbf{V} \mathbf{V}^H = \mathbf{I}_t$$

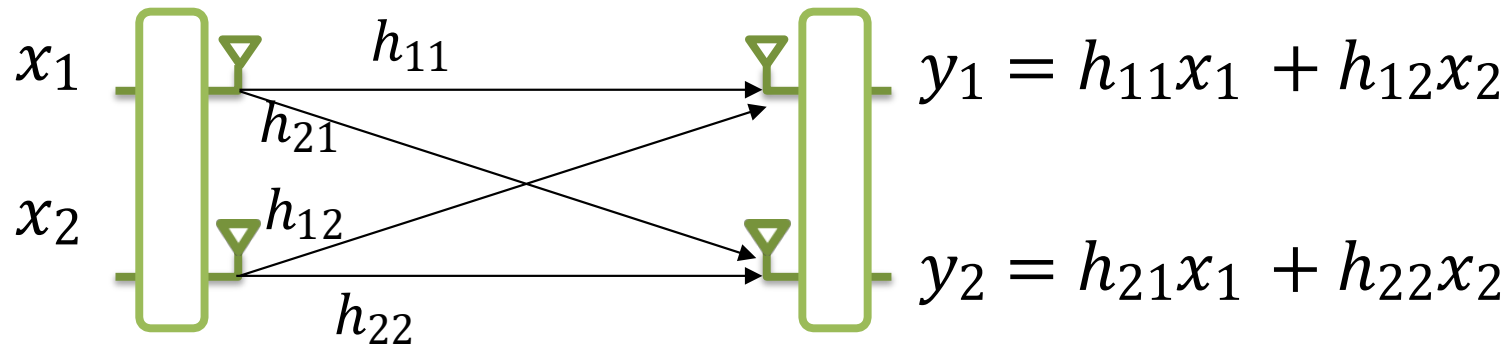
$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_t$$

$$\underbrace{\begin{bmatrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \\ v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{bmatrix}}_{\mathbf{V}^H},$$

$\left. \begin{matrix} v_{11}^* & v_{21}^* & \dots & v_{t1}^* \\ v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{matrix} \right\} \mathbf{v}_1^H$
 $\left. \begin{matrix} v_{12}^* & v_{22}^* & \dots & v_{t2}^* \\ \vdots & \vdots & \ddots & \vdots \\ v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{matrix} \right\} \mathbf{v}_2^H$
 $\left. \begin{matrix} v_{1t}^* & v_{2t}^* & \dots & v_{tt}^* \end{matrix} \right\} \mathbf{v}_t^H$

$$= \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H$$

SVD-based MIMO



$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

$$\tilde{\mathbf{y}} = \mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \tilde{\mathbf{n}}$$

$$\mathbf{y} = \underbrace{\mathbf{U}\mathbf{\Sigma}\mathbf{V}^H}_{\mathbf{H}}\mathbf{x} + \mathbf{n}$$

$$= \mathbf{\Sigma}\mathbf{V}^H\mathbf{V}\tilde{\mathbf{x}} + \tilde{\mathbf{n}}$$

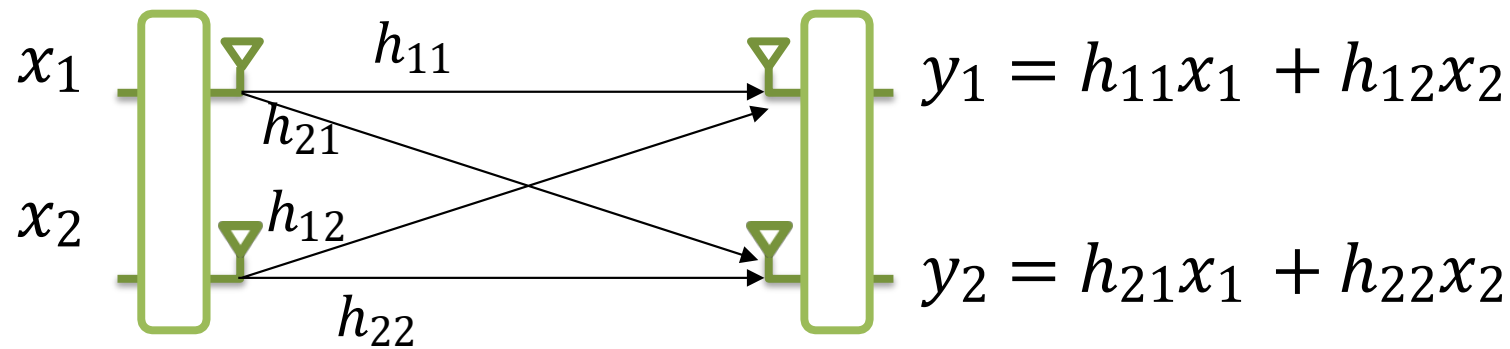
$$\underbrace{\mathbf{U}^H}_{\tilde{\mathbf{y}}}\mathbf{y} = \mathbf{U}^H(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \mathbf{n})$$

$$= \mathbf{\Sigma}\tilde{\mathbf{x}} + \tilde{\mathbf{n}}$$

$$\tilde{\mathbf{y}} = \underbrace{\mathbf{U}^H\mathbf{U}}_{\mathbf{I}_t}\mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \underbrace{\mathbf{U}^H}_{\tilde{\mathbf{n}}}\mathbf{n}$$

$$= \mathbf{\Sigma}\mathbf{V}^H\mathbf{x} + \tilde{\mathbf{n}}$$

SVD-based MIMO



$$\begin{bmatrix} \tilde{y}_1 \\ \tilde{y}_2 \\ \vdots \\ \tilde{y}_t \end{bmatrix} = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \vdots \\ \tilde{x}_t \end{bmatrix} + \begin{bmatrix} \tilde{n}_1 \\ \tilde{n}_2 \\ \vdots \\ \tilde{n}_t \end{bmatrix}$$

$$\tilde{y}_1 = \sigma_1 \tilde{x}_1 + \tilde{n}_1$$

$$\tilde{y}_2 = \sigma_2 \tilde{x}_2 + \tilde{n}_2$$

$$\vdots$$

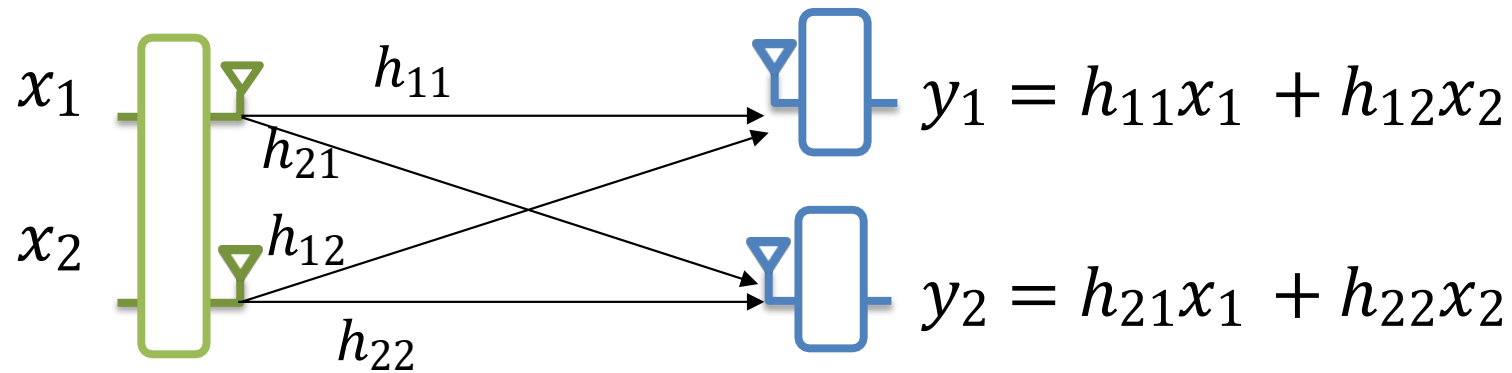
$$\tilde{y}_t = \sigma_t \tilde{x}_t + \tilde{n}_t$$

$$C_i = \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_n^2} \right)$$

Total Capacity

$$C = \sum_{i=1}^t \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_i^2} \right)$$

Multi-User MIMO

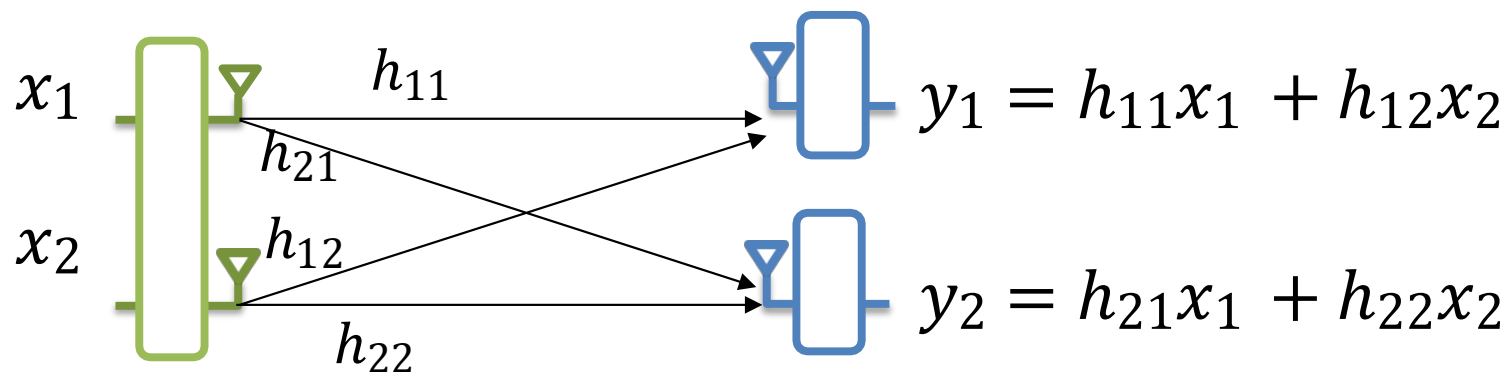


$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

$$\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y} = \mathbf{H}^{-1}\mathbf{H}\mathbf{x} + \mathbf{H}^{-1}\mathbf{n} = \mathbf{x} + \mathbf{H}^{-1}\mathbf{n}$$

Does not work. Receivers do not have access to each other's signals

Precoding



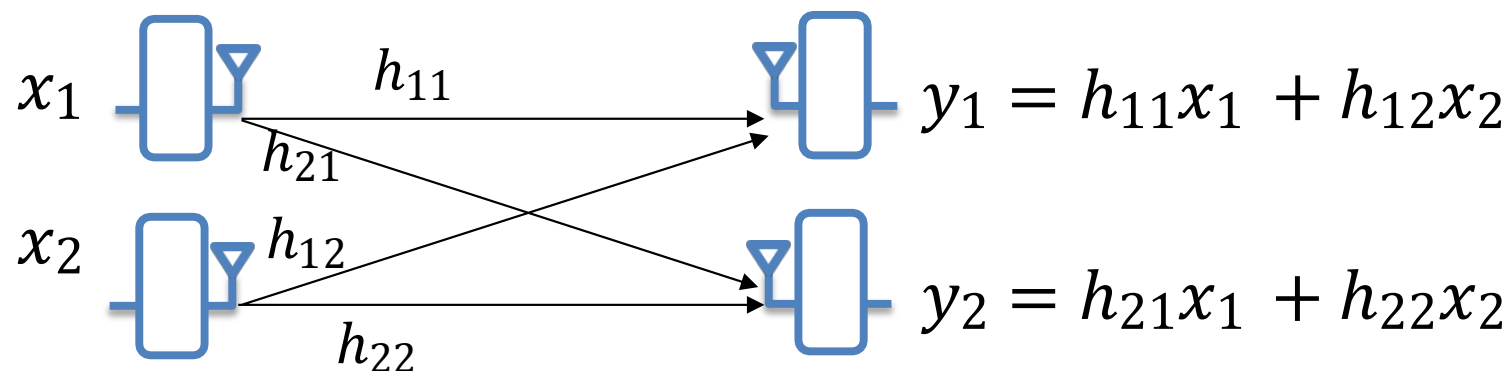
Send: $\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{x}$

Receive: $\mathbf{y} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n} = \mathbf{H}\mathbf{H}^{-1}\mathbf{x} + \mathbf{n} = \mathbf{x} + \mathbf{n}$

Also known as Beamforming or Zero-Forcing

Requires Channel Feedback!

What about distributed transmitters?



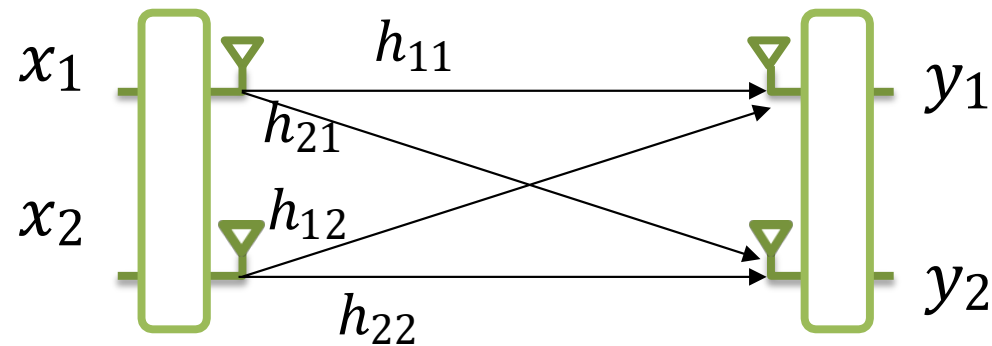
Send: $\tilde{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{x}$

Receive: $\mathbf{y} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n} = \mathbf{H}\mathbf{H}^{-1}\mathbf{x} + \mathbf{n} = \mathbf{x} + \mathbf{n}$

Does not work. Transmitters do not have access to each other's signals

False: what if transmitters are APs connected over Ethernet?

Clock Synchronization in MIMO

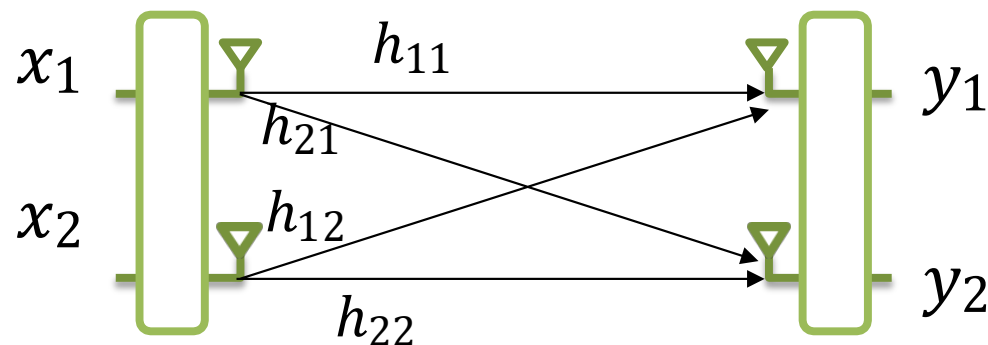


$$y_1 = h_{11}x_1e^{-j2\pi\Delta f_{11}t} + h_{12}x_2e^{-j2\pi\Delta f_{12}t}$$

$$y_2 = h_{21}x_1e^{-j2\pi\Delta f_{21}t} + h_{22}x_2e^{-j2\pi\Delta f_{22}t}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO

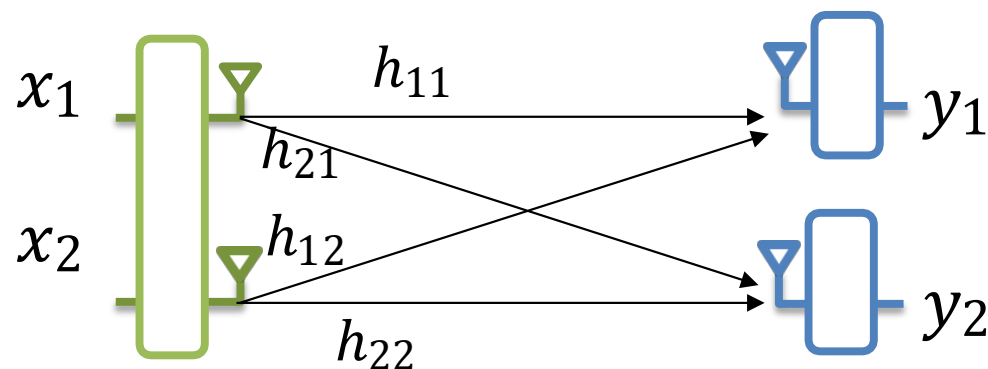


$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

On chip MIMO: $\Delta f_{11} = \Delta f_{21} = \Delta f_{12} = \Delta f_{22} = \Delta f$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} e^{-j2\pi\Delta f t} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO

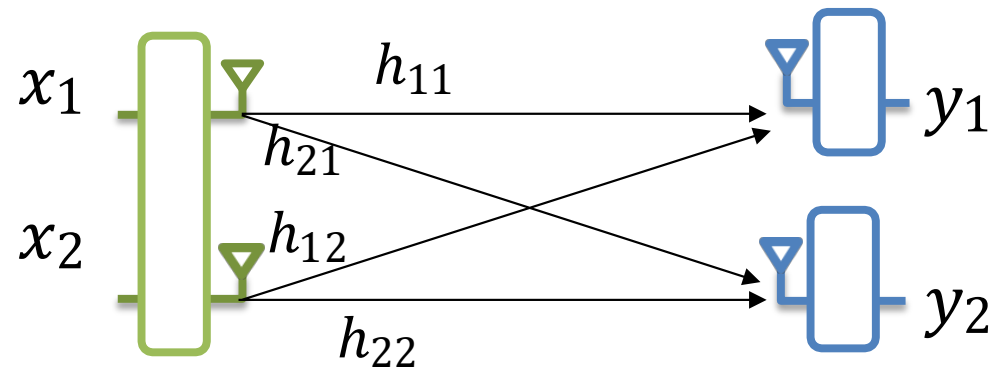


$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

MU-MIMO: $\Delta f_{11} = \Delta f_{12}$ and $\Delta f_{21} = \Delta f_{22}$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_1t} & h_{12} e^{-j2\pi\Delta f_1t} \\ h_{21} e^{-j2\pi\Delta f_2t} & h_{22} e^{-j2\pi\Delta f_2t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO

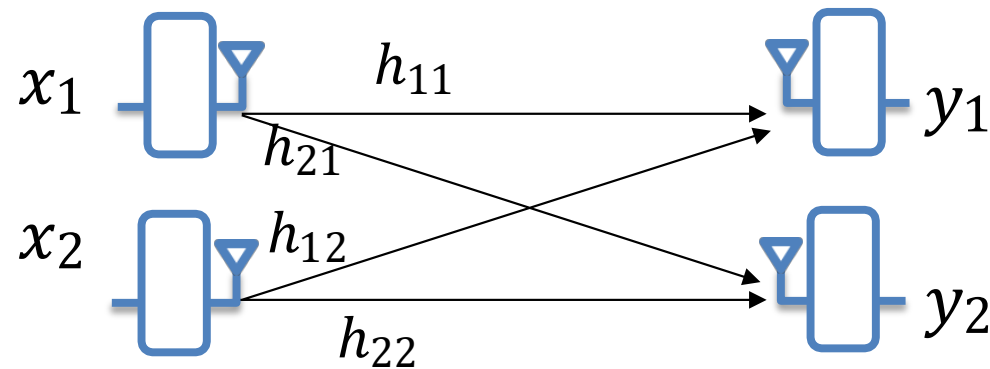


$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

MU-MIMO: $\Delta f_{11} = \Delta f_{12}$ and $\Delta f_{21} = \Delta f_{22}$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} e^{-j2\pi\Delta f_1 t} & 0 \\ 0 & e^{-j2\pi\Delta f_2 t} \end{bmatrix} \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Clock Synchronization in MIMO



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} h_{11} e^{-j2\pi\Delta f_{11}t} & h_{12} e^{-j2\pi\Delta f_{12}t} \\ h_{21} e^{-j2\pi\Delta f_{21}t} & h_{22} e^{-j2\pi\Delta f_{22}t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

Distributed MIMO: $\Delta f_{11} \neq \Delta f_{12} \neq \Delta f_{21} \neq \Delta f_{22}$